

Course code :- ITA0502

Course :- Computer vision

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ITA0502- COMPUTER VISION

MODEL LAB PRACTICALS

SET 8

1. Write a Python function to create a white image size entered by the user and then create a shape of Circle using Open CV.
2. Write a Python function to create a text string entered by the user that must be appeared on the given image using Open CV.
3. Write a Python function to subtract the background of the given input image based on color levels using Open CV
4. Write a Python function to subtract the foreground of the given input image based on color levels using Open CV.

Model Lab

Aim :- To create a white image size entered by user & then create a shape of circle using open cv.

Procedure :-

- (i) Import the open cv
- (ii) Get the image height & width
- (iii) Create a white image using numpy
- (iv) calculate image centre
- (v) Draw a circle using open cv
- (vi) Show the image.

Program :-

```
import cv2
import numpy as np
height = int(input())
width = int(input())
img = np.ones((height, width, 3))
centre = (width // 2, height // 2)
radius = min((width, height) // 2)
cv2.circle(img, centre, radius, (0, 0, 255), 3)
cv2.imshow("circle image", img)
cv2.imwrite("circle img.png", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Input :- image width : 600
image height : 400

Output :- white image.

Result :- Created successfully a white image using open cv.

2

Aim :- To create a text string entered user that must be appeared image using

Procedure :-

- (i) import open cv
- (ii) create a text string
- (iii) Give input as image
- (iv) centre the text string to the
- (v) Show the image.

Program :-

```
import cv2
import numpy as np
img = np.zeros(100, 100, 3)
text = input("text: ")
cv2.putText(img, text, (50, 200), cv2.FONT_HERSHEY_SIMPLEX,
              1.5, (0, 0, 0), 3)
cv2.imshow("Text image", img)
cv2.show()
```

Sample Input :- Enter Text : Nitheesh
Image : "img"

Output :- Text with image.

Result :- Successfully created a text string appeared as image.

Aim :- To create a python function to subtract the background of input image based on colour levels.

Procedure :-

- (i) Import opencv.
- (ii) Read the input image.
- (iii) convert the image RGB to HSV colour space.
- (iv) create a mask using cv2.inRange()
- (v) apply mask to original image.
- (vi) display the background.

Program :-

```
import cv2
import numpy as np
img = cv2.imread("input.jpg")
hsv = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)
lower = np.array([0, 0, 180])
upper = np.array([180, 255, 255])
background_mask = cv2.inRange(hsv, lower, upper)
foreground_mask = cv2.bitwise_not(background_mask)
result = cv2.bitwise_and(img, img, mask=foreground_mask)
cv2.imshow("Background subtracted")
```

Input :- Input Image : input.jpg.
Background :- white space image.

Output :- Background is removed from image.

4. Aim :- To convert subtract to the foreground of the image based on colour levels using opencv.

Procedure :-

- (i) import opencv
- (ii) Read the input image.
- (iii) convert the image HSV to RGB
- (iv) create a mask using opencv
- (v) Apply mask to the original image.
- (vi) Display the foreground image.

Program :-

```
import cv2
import numpy as np
img = cv2.imread("input.jpg")
hsv = cv2.cvtColor(img, cv2.COLOR_RGB2HSV)
lower = np.array([0, 50, 50])
upper = np.array([180, 255, 255])
foreground_mask = cv2.inRange(hsv, lower, upper)
background_mask = cv2.inRange(hsv, lower, upper)
result = cv2.bitwise_and(img, img, mask=background_mask)
cv2.imshow("original image", img)
cv2.imshow("foreground subtracted", result)
cv2.destroyAllWindows()
```

Input :- Input :- Image.jpg

foreground :- specified person

Output :- specified person removed from image.